PH-UY 2023: ELECTRICITY, MAGNETISM, & FLUIDS

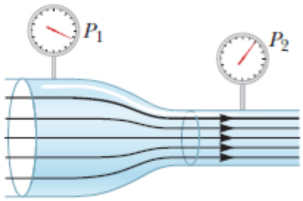
FINAL EXAM
12/17/25

Name _____
NetID _____

BEFORE YOU START…

On the exam, blue book, and answer sheet, write:
- your FULL name (as listed by the university registrar)
- your NetID (i.e. your NYU email minus "@nyu.edu")

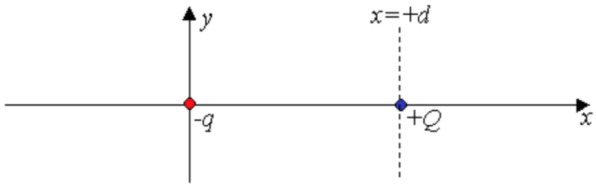
1. Water flows through a pipe whose cross-sectional area gradually tapers to 2/3 its original value, as shown in the figure. The pressure P_2 at the center of the narrow section is 0.055 atm lower than the pressure P_1 at the center of the wider section. What is the speed of the water at the center of the *wider* section of the pipe?



a) 1.27 m/s b) 2.01 m/s c) 2.38 m/s d) 2.70 m/s e) 2.99 m/s

Questions 2 and 3 refer to the following situation.

A point particle of positive charge +Q is fixed at rest at the point (*d*,0), while another negative charge *-q* = -8Q is fixed at rest at the origin.



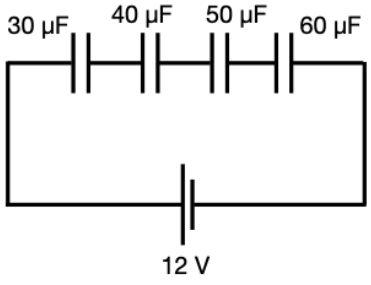
2. What is the x-coordinate of the point on the x-axis where the electric field vanishes?

a) +1.43 d b) +1.55 d c) +1.69 d d) +1.81 d
e) There is no point on the x-axis at which the electric field vanishes.

3. A third particle of charge +19 *Q* initially at rest very far from the origin is now brought to a point P on the positive part of the **y-axis** (note: the y-axis, *NOT* the x-axis) and affixed there at rest. If zero work was done to bring in that third particle, what is the y-coordinate of point P?

a) +1.43 d b) +1.55 d c) +1.69 d d) +1.81 d
e) There is no point P on the +y-axis that satisfies all the conditions stated in the problem.

4. In the circuit shown, what is the voltage across the 30 μF capacitor?



a) 4.21 V b) 2.11 V c) 3.16 V
d) 2.53 V e) 12.0 V

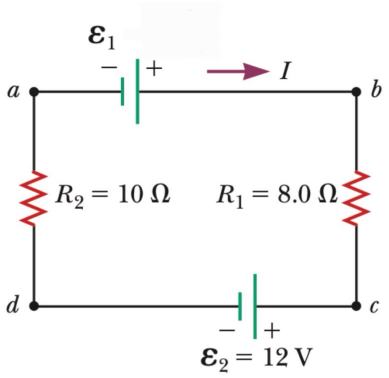
5. A 15 μF parallel plate capacitor with no dielectric between the plates is connected to a 12-volt battery and given sufficient time to fully charge. The battery is then disconnected and a material with dielectric constant of 4.3 is inserted between the capacitor plates. How much stored energy is lost when the dielectric is added?

a) 909 μJ b) 0 μJ c) 829 μJ d) 753 μJ e) 610 μJ

6. Two wires have the same resistance and the same length. One is made from copper and the other from aluminum. Given that the resistivity of copper is 1.68×10^{-8} Ω·m and that of aluminum is 2.70×10^{-8} Ω·m, what is the ratio of the radius of the copper wire to the radius of the aluminum wire?

a) 0.345 b) 0.789 c) 0.830 d) 0.455 e) 0.490

7. A single-loop circuit contains two resistors and two batteries as shown in the figure. Find the current in the circuit if the top battery has a voltage of $\varepsilon_1 = 4$ V.



a) -0.222 A b) -0.111 A c) -0.555 A d) -0.333 A e) -0.444 A

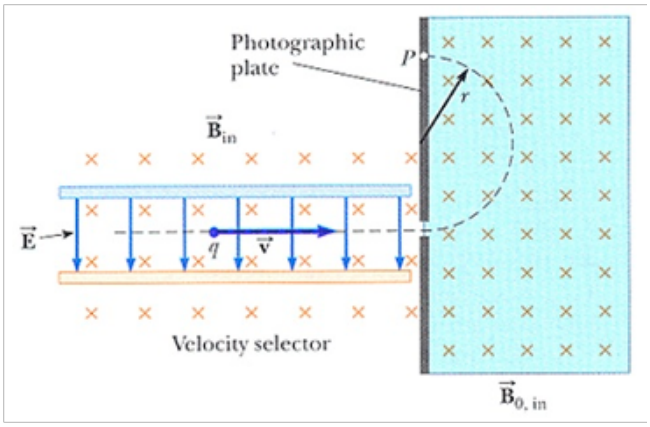
8. An initially uncharged 20 μF capacitor is connected in series with a 10 V battery and a resistor with resistance R. 3 seconds after it begins charging, the potential difference between the capacitor plates has grown to 4.00 V. What is the value of R?

a) 4.89×10^5 Ω b) 5.87×10^5 Ω c) 6.87×10^5 Ω
d) 3.92×10^5 Ω e) 2.94×10^5 Ω

9. A horizontal wire of linear mass density 0.05 kg/m carries a current of 4.9 A. If the wire is immersed in a uniform external magnetic field of magnitude B, what is the minimum value of B needed for the wire to levitate above the ground?

a) 0.12 T b) 0.16 T c) 0.14 T d) 0.20 T e) 0.10 T

10. An atom of mass 2.18×10^{-26} kg and charge 1.6×10^{-19} C passes through a velocity selector into a rectangular deflection chamber as shown in the figure below. In the velocity selector, a uniform magnetic field of magnitude B_{in} points into the page, and a uniform electric field of magnitude E points down. In the chamber, a separate uniform magnetic field of magnitude $B_{0,in} = 0.34$ T (note: $B_{0,in} \neq B_{in}$) bends the ion along a semicircular arc of radius r = 20 mm until it strikes a plate at P. If $E = 2.70 \times 10^4$ V/m, what is the magnitude of the magnetic field B_{in} within the velocity selector?



a) 0.541 T b) 0.361 T c) 0.381 T d) 0.481 T e) 0.421 T

11. Four long wires, parallel to the z-axis, pass through the corners of a square of side 4.0 cm in the xy-plane. Each wire carries 20 A in the same direction. What is the magnitude of the net magnetic force per meter of its length on any one wire due to the other three?

a) 7.17 mN b) 8.92 mN c) 9.83 mN d) 4.24 mN e) 5.61 mN

12. A rectangular loop of wire carrying a 4.0-A current is placed in a magnetic field of 0.60 T. The magnitude of the torque acting on this wire when the *plane of the loop* makes a 20° angle with the field is measured to be 1.1 N·m. What is the area of this loop?

a) 1.08 m^2 b) 0.488 m^2 c) 0.560 m^2 d) 0.648 m^2 e) 1.43 m^2

13. A conducting bar slides to the right on frictionless, conducting rails. The rails are connected by a 4.00-Ω resistor, forming a rectangular circuit. The bar has length 1.60 m and moves to the right at a constant speed of 2.1 m/s. The region contains a uniform 2.20-T magnetic field directed perpendicular to the rectangular circuit. What is the power dissipated in the 4.00-Ω resistor?

a) 15.4 W b) 29.8 W c) 24.3 W d) 17.8 W e) 13.7 W

14. A circular loop of conducting wire has radius 2.0 m and lies in a uniform magnetic field perpendicular to the plane of the loop. The magnetic field then begins to change in time. If the maximum induced emf in the loop is 5 V, what is the maximum rate at which the magnetic field strength is changing?

a) 0.398 T/s b) 0.557 T/s c) 0.477 T/s d) 0.637 T/s e) 0.211 T/s

15. A solenoid with length 15 cm and cross-sectional area 1 cm^2 has an inductance of 1.20 mH. When the wire windings carry a current of 2.5 A, what is the energy density in the middle of the solenoid?

a) 625 J/m^3 b) 750 J/m^3 c) 250 J/m^3 d) 375 J/m^3 e) 500 J/m^3